

Global Question-Aware Multimodal Retrieval-Augmented Generation for Multimedia Multi-Hop Question Answering

Zhixiao Shen

Nanjing University of Science and Technology
Nanjing, China
zxshen@njjust.edu.cn

Wenya Wang

Nanyang Technological University
Singapore, Singapore
wangwy@ntu.edu.sg

Jianfei Yu*

Nanjing University of Science and Technology
Nanjing, China
University of Chinese Academy of Sciences
Nanjing, China
jfyu@njjust.edu.cn

Rui Xia

Nanjing University
Nanjing, China
University of Chinese Academy of Sciences
Nanjing, China
rxia.cn@gmail.com

Abstract

Multimedia Multi-Hop Question Answering (MMQA) is a complex task that requires models to reason over and integrate information from both visual (e.g., images) and textual (e.g., documents) modalities to answer questions that cannot be resolved in a single step. Existing methods for MMQA suffer from two common limitations: (1) insufficient cross-modal information fusion, which restricts interaction between different modalities; and (2) weak global understanding of multi-hop questions, making them vulnerable to distractions from intermediate steps. To address these challenges, we introduce Global question-aware Multimodal Retrieval-Augmented Generation (GMRAG), a framework designed to enhance cross-modal reasoning and improve retrieval precision for multi-hop multimodal questions. It brings two core innovations: (1) restructuring the training data into a global question-aware, evidence-centric retrieval dataset, enabling the retriever to perform richer cross-modal reasoning on multi-hop multimodal questions and to identify globally relevant information more effectively; and (2) employing cross-modal contrastive learning to fine-tune a joint image-text encoder, achieving stronger alignment between holistic multimodal questions and evidence. The retrieved evidence is then directly fed into a Multimodal Large Language Models (MLLMs) to generate the final answer to the original question, eliminating dependence on potentially flawed intermediate answers and thus mitigating error propagation. Experimental results on two public MMQA datasets show that GMRAG consistently outperforms existing RAG methods across various MLLMs.

*Corresponding author.

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CCS Concepts

• **Computing methodologies** → **Natural language processing**; **Natural language processing**; • **Information systems** → **Information retrieval**.

Keywords

Multimodality, Vision Question Answering (VQA), Retrieval Augmented Generation (RAG)

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1 Introduction

Understanding complex information in real-world scenarios (such as interpreting news articles, scientific reports, or social media posts) often requires the integration of multimodal sources like images and text. Humans naturally integrate information from multiple modalities to answer complex questions, drawing connections between images and text to build a comprehensive understanding of the world. Inspired by this human capability, the Multimedia Multi-Hop Question Answering (MMQA) task has emerged, requiring systems to answer multi-hop questions by reasoning over visual and textual modalities, such as an accompanying image and document [16]. In MMQA, models are given a question, an image, a caption and an accompanying document, and must perform multi-hop, cross-modal reasoning to derive the answer. As illustrated in Figure 1, answering a question like “What is the man holding the book doing?” first requires grounding the visual reference—the *man holding the book*—to *Nicolas Maduro* in the image, and then reasoning over the text to find the answer, such as *dismantling Venezuela’s democracy*. This highlights the need for both cross-modal grounding and multi-step textual inference.

Several prior studies have tackled MMQA by decomposing multi-hop questions into modality-specific sub-questions, solving them

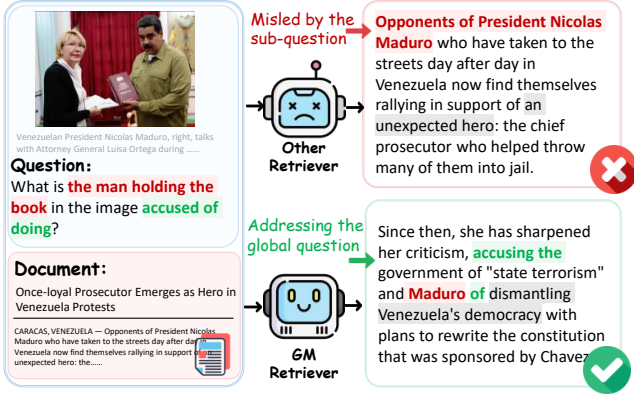


Figure 1: Comparison between the previous multimodal retriever and our Global question-aware Multimodal (GM) retriever. The text in gray shadow is the answer generated by MLLM based on the corresponding evidence.

in a stepwise manner [6, 16, 19]. This pipeline design inherently restricts reasoning to a single modality at each step, preventing deeper cross-modal semantic integration. Moreover, early errors in sub-question answering tend to propagate to later stages, ultimately compromising the final prediction.

To overcome these challenges, a promising approach is to retrieve and reason over evidence holistically, rather than fragmenting the question into independent sub-questions. Inspired by the recent success of Multimodal Retrieval-Augmented Generation (MRAG) in multimodal QA tasks [2, 18, 26], we consider an MRAG-based solution that retrieves relevant evidence from the accompanying document and integrates it into Multimodal Large Language Models (MLLMs) for answer generation. Existing MRAG approaches typically follow two directions: some use separate unimodal encoders for text and images [3, 13, 15], which limit cross-modal alignment and result in suboptimal evidence retrieval; others employ joint image-text encoders [5, 29, 30], which, as shown in Figure 1, tend to be distracted by intermediate sub-questions and fail to retrieve the key text chunks needed to answer the full multi-hop question.

To address these limitations, we propose Global question-aware Multimodal Retrieval-Augmented Generation (GMRAG), a framework that enhances cross-modal reasoning and improve retrieval precision for multi-hop multimodal questions. GMRAG aims to retrieve evidence that is globally relevant and semantically aligned with the entire multi-hop question. To achieve this, we restructure the training data into a global question-aware, evidence-centric retrieval dataset, where each question is paired with positive and hard negative evidence. This construction enables more effective supervision during retriever training. We then apply a cross-modal contrastive learning strategy to fine-tune a joint image-text encoder, aligning holistic questions with the most semantically relevant evidence segments across modalities. This optimization enhances the retriever’s ability to handle multi-hop, multimodal queries that require cross-modal reasoning. Finally, the retrieved context and question is fed into MLLMs for final answer generation.

To validate the effectiveness of GMRAG, we conduct extensive experiments on the MultimodalQA and MuMuQA benchmarks. Results show that GMRAG consistently outperforms existing RAG

baselines, with a 2.23% F1 improvement over VISTA, and better supports MLLMs in generating accurate answers.

Our main contributions can be summarized as follows:

- We propose a novel Global question-aware Multimodal Retrieval-Augmented Generation (GMRAG) framework for MMQA, which retrieves and integrates evidence with a holistic view of the question.
- We construct a global question-aware, evidence-centric retrieval dataset by reorganizing MMQA training data and apply cross-modal contrastive learning to fine-tune a joint image-text retriever, enhancing alignment between global questions and evidence while mitigating distractions from intermediate steps.
- We integrate the retrieved evidence into various MLLMs and demonstrate through experiments on MultimodalQA and MuMuQA datasets that GMRAG significantly outperforms existing RAG-based baselines across multiple evaluation settings.

2 Related Work

2.1 Multi-modal Multi-hop Question Answering

Several MMQA datasets have been introduced in recent years. For example, **MultimodalQA** includes text, tables, and images, covering both unimodal and multimodal questions [19], while **MuMuQA** focuses on cross-media reasoning in the news domain [16]. Early solutions used *modality-specific* architectures, where separate QA models were trained per modality and selected by a classifier [9, 19]. Later, *hierarchical decomposition* methods broke multi-hop questions into modality-grounded sub-questions [6, 16]. Recent work explores retrieval-augmented paradigms. Some unify multimodal content into a text format for retrieval [27], while others encode each modality separately and retrieve evidence via similarity-based methods [2, 13, 18, 26]. However, these still struggle with effective cross-modal reasoning.

2.2 Retrieval-Augmented Generation for QA

Retrieval-augmented generation (RAG) has shown promise in open-domain and multimodal QA. **GraphRAG** [4] integrate large-scale knowledge by building entity graphs across documents, but is inefficient for single-document QA due to graph sparsity and high computational cost. **Agentic RAG** [10] incorporates planning and web search for question decomposition, but its open-domain nature makes it unnecessary for document-grounded QA. **VisRAG** [28] integrates visual layouts by treating documents as images and using MLLMs for embedding and retrieval. However, it neglects textual parsing and lacks explicit cross-modal interaction, limiting its performance on fine-grained multimodal reasoning tasks. In related generation tasks, **RaPSG** [11] proposes retrieval-augmented pseudo-captioning using region-level descriptions from mismatched corpora. It leverages pretrained models for supervision, showing the benefit of retrieval-enhanced generation in low-resource settings.

2.3 Multimodal Retrievers and MLLMs

Marvel [30] uses visual plugins to project image features into the language space, while **VISTA** [29] enhances visual understanding

via visual tokens fused with a strong text encoder. Many task-specific cross-modal retrieval methods [7, 8, 14, 24, 25] focus on particular modality pairs, employing tailored feature modeling and alignment to improve benchmark performance. However, these methods mainly target single-hop QA and lack the generality and robust text retrieval strategies needed for multimodal queries. Modern **MLLMs** extend language models with visual encoders and fusion modules. For example, **LLaVA** [12] uses linear projection for visual token alignment, and **Qwen2-VL** [21] introduces position-aware adapters. Scaled systems like **Qwen-VL-max** [1] and **Gemini** [20] further enhance visual-language alignment. Nonetheless, empirical studies show that directly applying MLLMs to MMQA often underperforms [16, 19], highlighting the importance of integrating external retrievers for evidence grounding.

3 Method

3.1 Task Formulation and Method Overview

For the task of multimodal multi-hop question answering (MMQA), given a multimodal query consisting of a textual question q , a reference image v , and its corresponding caption c , the model is expected to generate a textual answer a by jointly reasoning over an accompanying document d , which contains textual passages. Unlike standard visual QA or single-hop multimodal reasoning, MMQA requires integrating evidence from multiple sources and performing multi-step reasoning across visual and textual modalities.

Previous methods often rely on document-only retrieval or shallow vision-text alignment, which limits their ability to capture global reasoning chains involving entities, events, and visual semantics. To address these challenges, we propose a new retrieval-augmented framework called GMRAG, as illustrated in Figure 2, which is designed to support global question-aware reasoning across modalities. Our framework consists of three key stages: (1) Retrieval Evidence Construction, which builds a training dataset of relevant and challenging evidence for the retriever (see Sec. 3.2); (2) Global Multimodal Retriever Training, which learns to retrieve semantically aligned document passages based on the fused representation of the question, image, and caption (see Sec. 3.3); and (3) GM Retrieval-Based Answer Generation, which uses the retrieved evidence along with the query to prompt a multimodal LLM to generate the final answer (see Sec. 3.4). During inference, given a multimodal query, our GM model first retrieves the top- P relevant textual chunks by computing similarity to the fused query representation. These retrieved chunks are then composed with the query image, caption, and the question into a structured prompt, which is fed into a pretrained MLLM to generate the answer. By leveraging fine-grained multimodal retrieval and MLLM’s generation, our framework enables robust multimodal multi-hop reasoning.

3.2 Retrieval Evidence Construction

To enhance the retriever’s ability to identify globally relevant evidence aligned with the semantics of multimodal multi-hop questions, we construct a retrieval dataset from the training corpus. This dataset contains both *positive* samples and *hard negatives*, specifically designed to supervise contrastive learning for the retriever.

3.2.1 Positive Evidence Construction. Given a training sample consisting of an image v , its caption c , a document d , a question q , and the ground-truth answer a , we begin by segmenting the document d into M non-overlapping text chunks $\{p_1, p_2, \dots, p_M\}$. We then extract the subset of chunks that contain the ground-truth answer span a as candidate positive evidence:

$$\mathcal{P}^+ = \{p_i \mid a \in p_i\}. \quad (1)$$

However, since multi-hop questions often require reasoning beyond localized textual spans, we further rank each $p_i \in \mathcal{P}^+$ based on its semantic relevance to the entire question context. This relevance is measured via similarity between the chunk representation e_{p_i} and a multimodal query representation e_q . We compute chunk embeddings using a BERT-based encoder:

$$e_{p_i} = \text{BERT}(t_0^p, t_1^p, \dots, t_{m_p}^p), \quad (2)$$

where $\{t_0^p \dots t_{m_p}^p\}$ is the tokenized sequence of p_i . To build the query embedding e_q , we incorporate visual, textual, and caption information using a cross-modal encoding scheme:

$$\begin{aligned} \{\xi_0, \dots, \xi_n\} &= \text{ViT}(i_0, i_1, \dots, i_m), \\ e_q &= \text{BERT}([\xi_0, \dots, \xi_n], [t_0^q, \dots, t_{m_q}^q], [t_0^c, \dots, t_{m_c}^c]), \end{aligned} \quad (3)$$

where $\{t_0^q \dots t_{m_q}^q\}$ and $\{t_0^c \dots t_{m_c}^c\}$ are tokens from the question and caption, respectively, while $\{i_0 \dots i_m\}$ is the token sequence of the input image patches. The sequence $\{\xi_0 \dots \xi_n\}$ corresponds to the sequence of hidden states for image tokens produced by ViT. The similarity score between chunk p_i and the multimodal query is computed via dot product: $\text{rel}_i = e_{p_i} \cdot e_q$. The chunk with the highest similarity is selected as the final positive evidence:

$$p^+ = \arg \max_{p_i \in \mathcal{P}^+} \text{rel}_i. \quad (4)$$

3.2.2 Hard Negative Evidence Construction. To encourage fine-grained discrimination during retriever training, we construct hard negative samples that are semantically similar to the positive evidence yet do not contain the correct answer. For each positive chunk p^+ , we identify negative chunks $p_j^- \notin \mathcal{P}^+$ from the same document. Each candidate negative is encoded using the same BERT-based encoder: $e_{p_j^-} = \text{BERT}(p_j^-)$, and its similarity to the positive chunk embedding is measured by the dot product: $s_j = e_{p_j^-} \cdot e_{p^+}$. We then select the top- K chunks with highest scores as hard negatives, denoted by $\mathcal{P}^-$.

To prevent overfitting or memorization, we randomly sample R hard negatives from $\mathcal{P}^-$ to form a negative sample set $\mathcal{N}$ for each positive chunk:

$$(p^+, \{p_j^-\}_{j=1}^R), \quad \{p_j^-\}_{j=1}^R \subseteq \mathcal{P}^-. \quad (5)$$

This evidence set provides strong supervision signals for training a multimodal retriever under contrastive learning, as described in Section 3.3.

3.3 GM Retriever Training

After constructing the contrastive supervision dataset, we train a global question-aware multimodal (GM) retriever that jointly models textual and visual inputs. Inspired by Zhou et al. [29], the GM retriever is built upon a dual-encoder architecture with cross-modal fusion, where both the vision transformer (ViT) and text

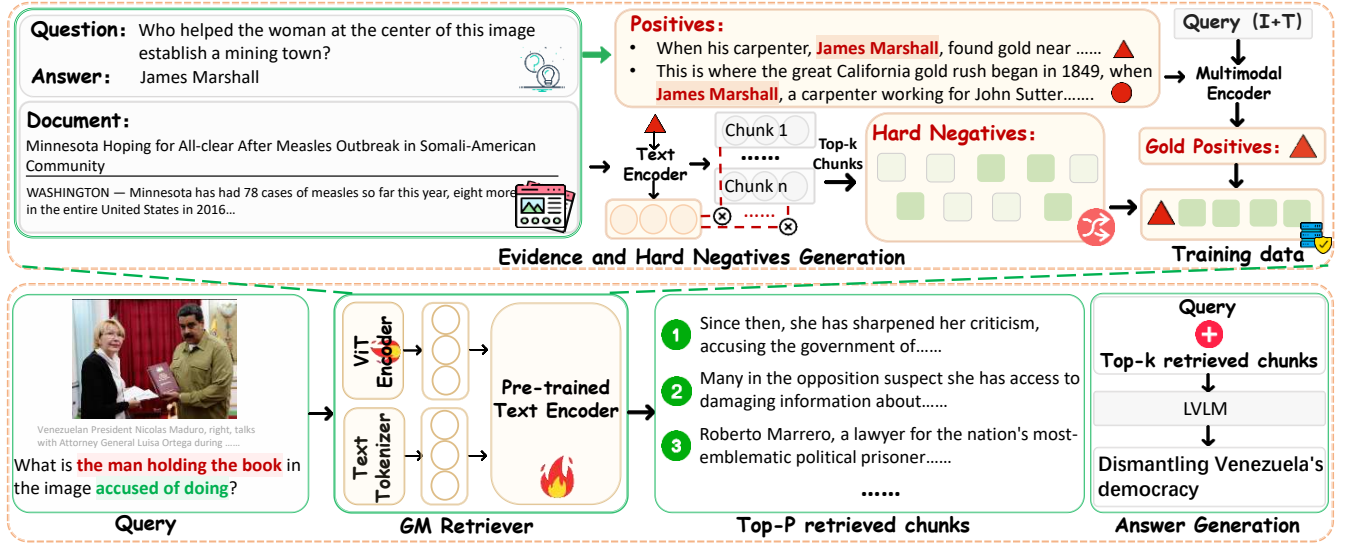


Figure 2: Overview of the proposed Global question-aware Multimodal Retrieval-Augmented Generation (GM-RAG) framework.

encoder are initialized from pretrained models and jointly fine-tuned for the retrieval task. We treat the ViT as an image tokenizer of the text encoder, which allows the pre-trained text model to recognize image tokens. The benefit of this approach is that it facilitates an in-depth fusion of text and image data.

Architecture Overview. The GM retriever consists of two primary modules:

- **Vision Transformer:** Given an input image v , the ViT encodes it into a sequence of patch-level embeddings $\{\xi_0, \dots, \xi_n\}$, which serve as visual tokens.
- **Text Encoder:** The text encoder is responsible for encoding the question q and the image caption c . To enable cross-modal interaction, we concatenate the visual tokens from the ViT with the tokenized question and caption sequences $[\xi_0, \dots, \xi_n, t_1^q, \dots, t_{m_q}^q, t_1^c, \dots, t_{m_c}^c]$, and feed this combined sequence into the text encoder.

To enable effective cross-modal reasoning, we adopt an early fusion mechanism, where visual tokens from ViT are concatenated with textual tokens and jointly encoded by the text encoder. This yields a multimodal query representation $\mathbf{e}_q$ that is sensitive to both the question intent and visual context.

Contrastive Training Objective. To supervise retrieval learning, we adopt a contrastive loss that maximizes the similarity between the query representation $\mathbf{e}_q$ and its corresponding positive evidence embedding $\mathbf{e}_{p_i^+}$ while pushing apart the similarities with hard negative evidences $\{\mathbf{e}_{p_j^-}\}$. The training objective is defined as:

$$\mathcal{L}_{\text{GM}} = -\log \frac{\exp(\mathbf{e}_q^\top \mathbf{e}_{p_i^+} / \tau)}{\sum_{j \in \mathcal{B}} \exp(\mathbf{e}_q^\top \mathbf{e}_{p_j} / \tau)}, \quad (6)$$

where τ is a temperature hyperparameter, and $\mathcal{B}$ refers to the set of in-batch samples belonging to the same example. This objective encourages the retriever to align semantically relevant multimodal queries with their ground-truth evidences.

Inference Stage. During inference, the retriever encodes the given question q , image v , and caption c into a unified embedding $\mathbf{e}_q$ using the same cross-modal encoding mechanism as in training. Each chunk p_i from the document is independently encoded into $\mathbf{e}_{p_i}$. We then compute the dot-product similarity $\text{Sim}_i = \mathbf{e}_q \cdot \mathbf{e}_{p_i}$ for all chunks, and retrieve the top- P chunks with the highest similarity scores as the final evidence set:

$$P_k^* = \text{Top-}P(\{\text{Sim}_i\}_{i=1}^M). \quad (7)$$

Cross-Modal Reasoning Capability. By integrating visual tokens directly into the text encoding process, the GM retriever achieves deeper semantic alignment between image and question. This design enables the retriever to attend to question-relevant visual cues when computing relevance scores. In addition, training under contrastive supervision with hard negatives sharpens the model's ability to distinguish between subtle semantic differences in the candidate chunks, further enhancing its multi-hop and multimodal reasoning performance.

3.4 GM Retrieval-Based Answer Generation

After retrieving a set of top- P evidence chunks P_k^* using the GM retriever, we construct a multimodal prompt to guide the final answer generation. This prompt integrates multiple sources of information essential for complex reasoning: (1) the input image, its caption, and the multi-hop question, which together establish the core multimodal context; (2) the retrieved evidence P_k^* , which provides semantically relevant and globally grounded textual information; and (3) a task instruction that defines the answer generation objective.

The composed prompt is then fed into powerful MLLMs, which integrates both visual and textual cues in an end-to-end manner to generate a natural language answer. The inclusion of retrieved evidence serves as an external memory, alleviating the model's burden of long-context retention and improving its ability to reason across modalities with greater accuracy and interpretability. Qualitative results demonstrate that retrieved evidence significantly enhances the

Table 1: Comparison of MMQA results on two datasets. Bold and underlined fonts denote the best and second-best results.

Methods	MuMuQA		MultimodalQA	
	F1	EM	F1	EM
<i>Qwen2-VL-7B</i> [21]				
Direct [21]	25.72	14.54	<u>52.52</u>	<u>44.17</u>
Decomposition [16]	<u>26.40</u>	12.10	-	-
BM25 [17]	22.19	12.50	44.97	37.42
BGE [23]	23.68	15.08	37.97	31.13
Marvel [30]	25.61	15.97	39.63	33.28
UniRAG [18]	24.92	15.08	47.18	40.18
VISTA [29]	25.89	<u>16.15</u>	47.04	39.11
GMRAG (Ours)	28.12	17.31	54.65	46.01
<i>LLaVA-NeXT-8B</i> [12]				
Direct [12]	26.16	17.13	46.53	<u>37.58</u>
Decomposition [16]	28.04	9.72	-	-
BM25 [17]	23.45	17.24	42.72	34.05
BGE [23]	24.24	17.40	35.79	27.61
Marvel [30]	25.94	17.84	37.43	28.99
UniRAG [18]	26.32	<u>18.20</u>	<u>46.70</u>	36.81
VISTA [29]	26.15	18.02	45.03	35.58
GMRAG (Ours)	<u>27.38</u>	19.09	51.00	40.95
<i>Gemini-1.5-Flash</i> [20]				
Direct [20]	<u>32.55</u>	21.14	<u>39.56</u>	<u>31.90</u>
VISTA [29]	30.56	19.80	37.04	30.21
GMRAG (Ours)	33.10	<u>20.52</u>	43.85	36.66
<i>Qwen-VL-Max</i> [1]				
Direct [1]	40.92	28.55	<u>65.47</u>	<u>57.98</u>
VISTA [29]	39.38	26.32	<u>56.96</u>	<u>49.85</u>
GMRAG (Ours)	41.64	28.64	67.90	60.58

correctness of generated answers, especially in multi-hop scenarios where neither the image nor the question alone is sufficient.

4 Experiments

4.1 Experimental Setup

Datasets. We evaluate our method on two benchmark datasets, i.e., MuMuQA [16] and MultimodalQA [19]. MuMuQA contains 1,121 QA pairs in the test set, requiring cross-media object alignment from news articles and images. Its training set is generated via data augmentation, from which we randomly select 3,795 samples to train our GM retriever. Since the test set lacks ground-truth (GT) evidence, we extract it using the method in Section 3.2 to evaluate the retrieval performance. MultimodalQA involves reasoning over text, tables, and images with 16 question templates. As our study focuses on vision-and-text QA, we select 652 “Compose(TextQ, ImageListQ)” samples and apply five-fold cross-validation.

Evaluation Metric. For evaluating our models, we follow the evaluation methods provided by the datasets, using F1 and EM as the main QA metrics. we use Recall@1/5/10 for MuMuQA and Recall@1/3/5 for Multimodal QA as the retrieval evaluation metrics, which is the same as the MRR metric.

Comparison Systems. We compare our method with several baselines: (1) *Direct*: feeding the entire document into MLLM prompts for answer generation. (2) *Decomposition* [16]: a pipeline-based approach that decomposes multi-hop questions into sub-questions, answered sequentially using LLMs. (3) *BM25* [17]: probabilistic text retriever based on term frequency and document length. (4) *BGE* [23]: BERT-based text retriever trained in three stages. (5) *Marvel* [30]: multimodal retriever using visual plugin for image-to-text projection. (6) *UniIR* [22]: instruction-guided unified multimodal retriever. (7) *VISTA* [29]: multimodal retriever combining strong text encoder. These retrievers retrieve chunks that are then fed into prompts of MLLMs (Table 1) for answer generation.

4.2 Main Results

In Table 1, we show the performance of different methods on the MMQA task. First, it is clear that the proposed GMRAG method achieves consistent improvements over all the compared systems across different MLLMs. Text-based RAG methods (BM25, BGE) show a decline compared to the Direct approach, while MRAG methods (Marvel, UniRAG, VISTA) perform better than text-based ones but only achieve slight improvements over Direct. In contrast, GMRAG demonstrates significant improvements in both EM and F1 scores. For example, on the MuMuQA dataset with the Qwen2-VL-7B model, GMRAG outperforms VISTA by 2.23 percentage points in F1 and surpasses the Decomposition approach by 5.3 points in EM. Similar superiority is observed on the MultimodalQA dataset. Furthermore, closed-source MLLMs demonstrate superior performance on cross-modal reasoning tasks, even achieving high performance without retrieval enhancement. While VISTA caused a slight decline, GMRAG delivered modest gains, highlighting the framework’s effectiveness in complex multimodal tasks.

4.3 Multimodal Retrieval Performance

Table 2 shows the retrieval performance of different methods on MuMuQA and MultimodalQA datasets. First, we can find that text-based retrievers underperform multimodal approaches, indicating textual features is insufficient to capture the key information in multimodal tasks. Second, our GM retriever achieves a remarkable MRR@1 improvement of 11.24% points over VISTA on MuMuQA and attains near-perfect Recall@5 of 98.77% on MultimodalQA. These results validate GM’s effectiveness in evidence retrieval of multimodal multi-hop reasoning tasks.

4.4 Retrieval Effectiveness of GM with Different Modalities and Top-k Selections

In Figure 4, we compare the performance of the GM retriever when using multimodal queries versus unimodal queries (i.e., text-only and image-only) for retrieving evidence to support MMQA answer generation. As shown in Figure 4(a), the F1 scores of unimodal retrieval gradually improve with increasing values of P , eventually approaching the performance of the Direct method. This indicates

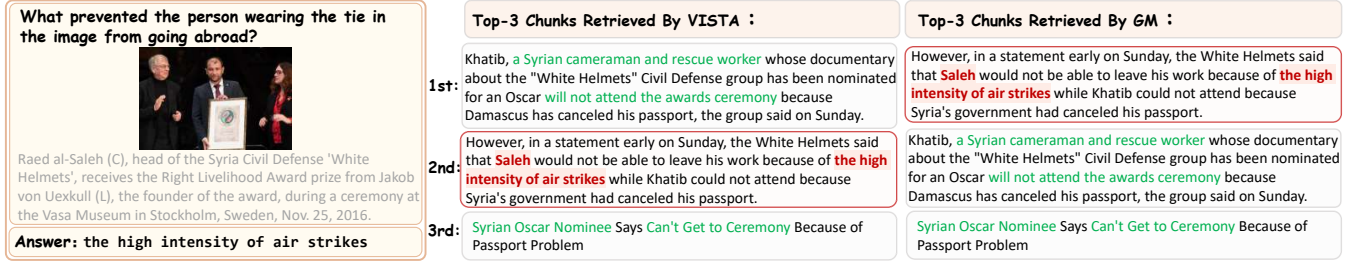


Figure 3: Top-3 Multimodal Retrieval cases from MuMuQA.

Table 2: Comparison of the retrieval performance.

MuMuQA	Methods	MRR @			Recall @		
		1	5	10	1	5	10
	BM25	13.38	25.60	28.82	13.38	49.60	73.95
	BGE	13.47	24.65	27.60	13.47	46.57	68.60
	MARVEL	12.76	29.07	32.26	12.76	59.59	83.05
	UniIR	16.68	30.51	33.26	16.68	55.75	76.36
	VISTA	15.34	34.27	36.95	15.34	67.43	87.24
	GM (Ours)	26.58	47.35	49.20	26.58	80.64	94.02

MultimodalQA	Methods	MRR @			Recall @		
		1	3	5	1	3	5
	BM25	38.50	51.46	55.19	38.50	68.10	84.20
	BGE	28.37	38.91	42.90	28.37	52.76	70.40
	MARVEL	28.22	39.85	44.36	28.22	56.29	76.38
	UniIR	45.40	58.13	60.87	45.40	74.54	86.81
	VISTA	49.39	59.94	62.49	49.39	73.77	84.82
	GM (Ours)	91.87	94.60	94.81	91.87	97.85	98.77

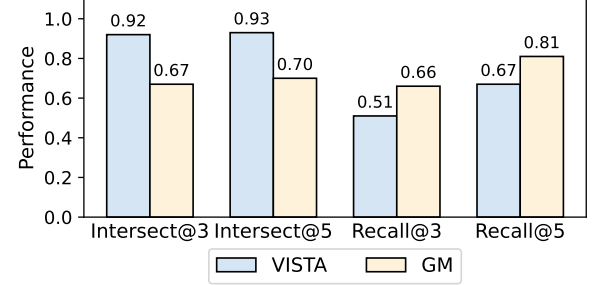


Figure 5: Comparison of GM and VISTA on Intersection and Recall Metrics.

Table 3: Sensitivity of model performance to the hyperparameter R

R	MRR@3	MRR@7	Recall@3	Recall@7
$R = 4$	44.72	48.96	66.90	87.03
$R = 8$	45.09	49.17	67.08	87.15
$R = 12$	44.99	49.17	67.17	87.33
$R = 16$	44.44	48.35	67.62	87.15

reasoning, whereas overly large P values may degrade the overall quality of the retrieved context.

4.5 Evaluating Whether GM Focuses on Global Question or Sub-Question

Figure 5 presents a comparison of the intersection and recall metrics for the GM and VISTA retrievers. The intersection metric evaluates the overlap between the chunks retrieved by sub-questions and those retrieved by the multi-hop question, while the recall metric evaluates the overlap between the chunks retrieved by the multi-hop question and the ground truth. The results show that GM achieves a significantly lower intersection score, indicating that it is less affected by interference from sub-questions. At the same time, GM outperforms VISTA on the recall metric, demonstrating its ability to retrieve paragraphs that effectively address the global question.

4.6 Sensitivity Analysis of Hyperparameter R

To assess the sensitivity of model performance to the hyperparameter R (the number of hard negatives for train GM), we conducted additional experiments by varying $R \in \{4, 8, 12, 16\}$. In Table 3, the results show that model performance is relatively stable across

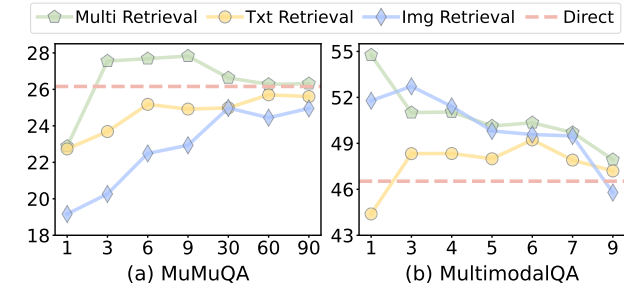


Figure 4: F1 scores of using GM retriever with different modalities and Top- P retrieved chunks on the MMQA task. We use LLaVA-NeXT-8B as the MLLM. (X-axis: Number of P , Y-axis: F1 score)

that unimodal queries provide limited utility in enhancing answer generation. In contrast, multimodal retrieval consistently outperforms unimodal retrieval across all P values and surpasses the Direct method in most settings. Notably, performance peaks when an appropriate P value is chosen (e.g., top-9). Figure 4(b) further shows that the effectiveness of multimodal retrieval decreases as P increases. Specifically, GM's recall reaches 91.87% at $P = 1$, and higher P values introduce noise, which negatively impacts performance. These results highlight that when P is optimally selected, multimodal retrieval can effectively support MLLMs in multi-hop

different values of R , indicating that our method is robust to this hyperparameter choice. While increasing R beyond 4 provides only marginal performance gains, it significantly increases computational cost due to the additional hard negative encoding required during training. Therefore, we select $R = 4$ as it strikes a good balance between effectiveness and efficiency.

4.7 Case Study

Figure 3 compares the top three retrieved chunks from VISTA and GM on a representative MuMuQA example. Based on the question, image, and caption, the query is about Saleh. Taking the top-ranked retrieved chunk as an example, VISTA retrieves a passage that primarily concerns another person (Khatib) and fails to identify the actual cause of Saleh’s inability to go abroad. In contrast, GM retrieves a passage that attributes the obstruction to air strikes, directly answering the question. This demonstrates GM’s stronger interaction capability across the question, caption, and image, enabling it to identify the core entity (“Saleh”) and retrieve the most relevant evidence. Rather than being misled by superficial lexical cues (e.g., “passport canceled”), GM grounds the retrieval in the actual query intent and is enabled by deeper cross-modal reasoning.

5 Conclusion

In this paper, we introduce a Global question-aware Multimodal Retrieval-Augmented Generation (GMRAG) framework to address the limitations of traditional retrieval methods that often fail to capture the global context in multimodal multi-hop questions. GMRAG consists of three key modules, i.e., Retrieval Evidence Construction, GM Retriever Training, and GM Retrieval-Based Answer Generation. Experimental results demonstrate that our GM retriever achieves superior retrieval performance in identifying semantically relevant evidence for multi-hop questions, which consequently leads to an absolute improvement in MMQA.

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